

this distance has three components:

Z_c = vertical deflection of the center-of-gravity;

$Z_{s_}$ = vertical deflection of a given spindle; and

H = vertical distance between the C.G. and the plane containing all four spindles at design position.

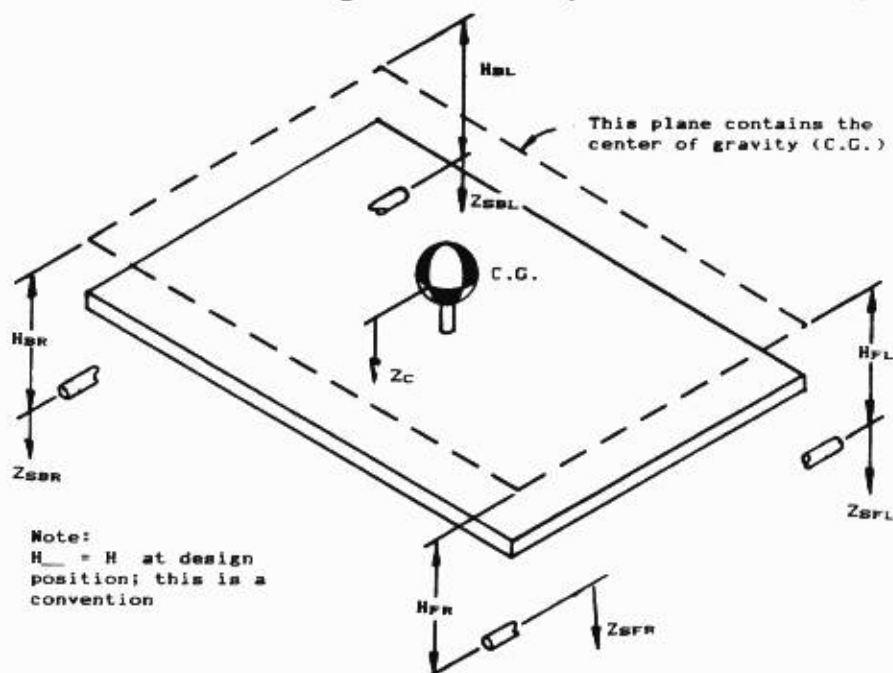


Figure 3-3: Center of gravity to spindle vertical distance

The first two of these parameters, Z_c and $Z_{s_}$, are calculated in the model. The H value is found by subtracting the static-loaded-radius of the tires from the vertical position of the C.G. relative to ground at design position. Consequently, the "C.G. to spindle distance" is defined for each spindle in Figure 3-3.

$$H_{FR} = H - Z_c + Z_{SFR} \quad (3-16)$$

$$H_{FL} = H - Z_c + Z_{SFL} \quad (3-17)$$

$$H_{BR} = H - Z_c + Z_{SBR} \quad (3-18)$$

$$H_{BL} = H - Z_c + Z_{SBL} \quad (3-19)$$

The subscripts used throughout this document will be explained as needed except for those used to express a specific corner